



JKBOSE 2026

CLASS 11TH

COMPLEX NUMBER

MATHS

LECTURE- 03

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TOPICS TO BE COVERED

1 **Multiplicative inverse of a Complex number**

2 **Exercise Discussion 4.1**

3 **Related Problems**

QUESTION

If $\left| \frac{z-2}{z-4} \right| = 1$, then $\text{Re}(z)$ is equal to :

1 3 ✓

2 0 ✗

3 -3 ✗

4 None of these ✗

(Sol) Given that $\left| \frac{z-2}{z-4} \right| = 1$

$$\Rightarrow \frac{|z-2|}{|z-4|} = 1$$

$$\Rightarrow |z-2| = |z-4|$$

$$\Rightarrow |x+iy-2| = |x+iy-4|$$

$$\Rightarrow |(x-2)+iy| = |(x-4)+iy|$$

$$\Rightarrow \sqrt{(x-2)^2 + y^2} = \sqrt{(x-4)^2 + y^2}$$

Squaring on b/s, we get;

$$\sqrt{(x-2)^2 + y^2}^2 = \sqrt{(x-4)^2 + y^2}^2$$

$$\Rightarrow (x-2)^2 + y^2 = (x-4)^2 + y^2$$

$$\Rightarrow x^2 - 4x + 4 = x^2 - 8x + 16$$

$$\Rightarrow -4x + 8x = 16 - 4$$

$$\Rightarrow 4x = 12$$

$$\Rightarrow x = 3$$

$$\Rightarrow \text{Re}(z) = 3$$

Multiplicative inverse of Z

let $z (\neq 0)$ be any non-zero Complex number,
then the multiplicative inverse of z (Reciprocal of z)
is denoted and defined as;

$$\bar{z}' = \frac{1}{z}$$

* We know that $z\bar{z} = |z|^2$
If $z \neq 0$, then

$$z = \frac{|z|^2}{\bar{z}}$$

$$\Rightarrow \frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

$$\Rightarrow \bar{z}' = \frac{\bar{z}}{|z|^2}$$

Q) Find the multiplicative inverse
of $z = 4 - 3i$.

(Sol) We know that

$$\bar{z}' = \frac{\bar{z}}{|z|^2}$$

$$\Rightarrow \bar{z}' = \frac{4 + 3i}{(\sqrt{4^2 + (-3)^2})^2}$$

$$\Rightarrow \bar{z}' = \frac{4 + 3i}{25}$$

$$\Rightarrow \bar{z}' = \left(\frac{4}{25}\right) + \left(\frac{3}{25}\right)i$$

EXAMPLE

Find the Complex number z for which $|z + 1| = z + 2(1 + i)$.

(Sol) Given that $|z + 1| = z + 2(1 + i)$

Put $z = x + iy$, we get,

$$|x + iy + 1| = (x + iy) + (2 + 2i)$$

$$\Rightarrow |(x + 1) + iy| = (x + 2) + i(y + 2)$$

$$\Rightarrow \sqrt{(x + 1)^2 + y^2} = (x + 2) + i(y + 2)$$

$$\Rightarrow \sqrt{(x + 1)^2 + y^2} + i \cdot 0 = (x + 2) + i(y + 2)$$

$$\text{So, } \sqrt{(x + 1)^2 + y^2} = (x + 2) \text{ \& } 0 = y + 2 \Rightarrow y = -2$$

$$\Rightarrow \sqrt{(x + 1)^2 + (-2)^2} = (x + 2)$$

$$\Rightarrow (x + 1)^2 + 4 = x^2 + 4x + 4$$

$$\Rightarrow x^2 + 2x + 1 = x^2 + 4x$$

$$\Rightarrow 2x - 4x = -1$$

$$\Rightarrow -2x = -1$$

$$\Rightarrow x = \frac{1}{2}$$

Therefore the required value of $z = x + iy$

$$\Rightarrow z = \frac{1}{2} - 2i$$

Hence the result.

EXAMPLE

Solve the equation $|z| + z = (2 + i)$ for a Complex value of z .

(Sol) Given that $|z| + z = 2 + i$

Put $z = x + iy$, we get,

$$|x + iy| + (x + iy) = 2 + i$$

$$\Rightarrow \sqrt{x^2 + y^2} + x + iy = 2 + i$$

$$\Rightarrow (\sqrt{x^2 + y^2} + x) + iy = 2 + i$$

So, $\sqrt{x^2 + y^2} + x = 2$ and $y = 1$

$$\Rightarrow \sqrt{x^2 + 1^2} + x = 2$$

$$\Rightarrow (\sqrt{x^2 + 1})^2 = (2 - x)^2$$

$$\Rightarrow x^2 + 1 = 4 + x^2 - 4x$$

$$\Rightarrow 4x = 4 - 1$$

$$\Rightarrow 4x = 3$$

$$\Rightarrow x = \frac{3}{4}$$

So the required Complex number is $z = x + iy$

$$\Rightarrow z = \frac{3}{4} + i$$

Hence the result.

QUESTION



If z satisfies the equation $z + \sqrt{2}|z + 1| + i = 0$, then z is equal to

☒ 1 $2 - i$

☐ 2 $-2 + i$

☒ 3 $-2 - i$

☐ 4 $2 + i$

(Sol) Given that $z + \sqrt{2}|z + 1| + i = 0$

Put $z = x + iy$, we get

$$(x + iy) + \sqrt{2}|x + iy + 1| + i = 0$$

$$\Rightarrow (x + iy) + \sqrt{2}|(x + 1) + iy| + i = 0$$

$$\Rightarrow x + iy + \sqrt{2}\sqrt{(x + 1)^2 + y^2} + i = 0$$

$$\Rightarrow (x + \sqrt{2}\sqrt{(x + 1)^2 + y^2}) + (y + 1)i = 0 + i0$$

$$\text{So } x + \sqrt{2}\sqrt{(x + 1)^2 + y^2} = 0 \text{ \& } y + 1 = 0 \\ \Rightarrow y = -1.$$

$$\Rightarrow x + \sqrt{2}\sqrt{(x + 1)^2 + 1} = 0$$

$$\Rightarrow \sqrt{2}\sqrt{(x + 1)^2 + 1} = -x$$

Squaring on b/s, we get,

$$(\sqrt{2}\sqrt{(x + 1)^2 + 1})^2 = (-x)^2$$

$$\Rightarrow 2((x + 1)^2 + 1) = x^2$$

$$\Rightarrow 2(x^2 + 2x + 1 + 1) = x^2$$

$$2(x^2 + 2x + 1 + 1) = x^2$$

$$\Rightarrow 2(x^2 + 2x + 2) = x^2$$

$$\Rightarrow 2x^2 + 4x + 4 - x^2 = 0.$$

$$\Rightarrow x^2 + 4x + 4 = 0.$$

$$\Rightarrow (x + 2)^2 = 0$$

$$\Rightarrow x + 2 = 0$$

$$\Rightarrow x = -2$$

So the required Complex number is,

$$z = -2 - 2i$$

EXAMPLE



If $z = 3 + 2i$, prove that $z^2 - 6z + 13 = 0$ and hence deduce $3z^3 - 13z^2 + 9z + 65 = 0$

(Sol) **M1**: Given that $z = 3 + 2i$

$$\Rightarrow z^2 = (3 + 2i)^2 = 3^2 + (2i)^2 + 12i$$

$$\Rightarrow z^2 = 9 - 4 + 12i$$

$$\Rightarrow z^2 = 5 + 12i$$

$$\text{Therefore } z^2 - 6z + 13 = (5 + 12i) - 6(3 + 2i) + 13$$

$$\Rightarrow z^2 - 6z + 13 = 5 + \cancel{12i} - 18 - \cancel{12i} + 13$$

$$\Rightarrow z^2 - 6z + 13 = \cancel{18} - \cancel{18} = 0$$

$$\Rightarrow z^2 - 6z + 13 = 0$$

$$\text{Further } z^3 = z^2 \cdot z = (5 + 12i)(3 + 2i)$$

$$\Rightarrow z^3 = 15 + 10i + 36i - 24$$

$$\Rightarrow z^3 = -9 + 46i$$

Now, Consider the following;

$$3z^3 - 13z^2 + 9z + 65$$

$$= 3(-9 + 46i) - 13(5 + 12i) + 9(3 + 2i) + 65$$

$$= -\cancel{27} + 138i - \cancel{65} - 156i + \cancel{27} + 18i + \cancel{65}$$

$$\Rightarrow 3z^3 - 13z^2 + 9z + 65 = \cancel{156i} - \cancel{156i}$$

$$\Rightarrow 3z^3 - 13z^2 + 9z + 65 = 0$$

M2: Given that $z = 3 + 2i$

$$\Rightarrow z - 3 = 2i$$

$$\Rightarrow (z - 3)^2 = (2i)^2$$

$$\Rightarrow z^2 + 9 - 6z = -4$$

$$\Rightarrow \boxed{z^2 - 6z + 13 = 0}$$

Further, Consider the following;

$$3z^3 - 13z^2 + 9z + 65$$

$$= 3z(z^2 - 6z + 13) + 5z^2 - 30z + 65$$

$$= 3z \times 0 + 5z^2 - 30z + 65.$$

$$\Rightarrow 3z^3 - 13z^2 + 9z + 65$$

$$= 5(z^2 - 6z + 13) = 5 \times 0$$

$$\Rightarrow \boxed{3z^3 - 13z^2 + 9z + 65 = 0}$$

Hence proved.

EXAMPLE

If $z = -5 + 3i$, then find the value of $z^4 + 9z^3 + 26z^2 - 14z + 8$

Homework Problem.

QUESTION

The number of solutions of the equation $z^2 + \bar{z} = 0$ is

- 1 ☒
- 2 ☒
- 3 ☒
- 4 ☒

(Sol) Given that $z^2 + \bar{z} = 0$

Put $z = x + iy$, we get,

$$(x + iy)^2 + \overline{x + iy} = 0$$

$$\Rightarrow x^2 - y^2 + 2xyi + x - iy = 0$$

$$\Rightarrow (x^2 - y^2 + x) + (2xy - y)i = 0 + i \cdot 0$$

$$\text{So } x^2 - y^2 + x = 0 \text{ \& } 2xy - y = 0 \quad \text{--- (1)}$$

$$\text{Here } 2xy - y = 0$$

$$\Rightarrow (2x - 1)y = 0$$

$$\Rightarrow x = \frac{1}{2}, y = 0$$

Put $y = 0$ in (1), we get

$$x^2 + x = 0$$

$$\Rightarrow x(x + 1) = 0$$

$$\Rightarrow x = 0, -1$$

$$\text{So, } z = 0, -1$$

Put $x = \frac{1}{2}$ in (1), we get

$$\frac{1}{4} - y^2 + \frac{1}{2} = 0$$

$$\Rightarrow y^2 = \frac{3}{4} \Rightarrow y = \pm \frac{\sqrt{3}}{2}$$

$$\text{So, } z = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \frac{1}{2} - i\frac{\sqrt{3}}{2}$$

QUESTION

$$\sqrt{-64} = \sqrt{64 \times -1} = 8i$$

$$|z^n| = |z|^n \quad |z^2| = |z \times z| = |z| \times |z| = |z|^2$$

If z is a complex number satisfying the equation $z^6 - 6z^3 + 25 = 0$, then the value of $|z|$ is equal to

(Sol) Given that $z^6 - 6z^3 + 25 = 0$.

$$\Rightarrow (z^3)^2 - 6z^3 + 25 = 0$$

$$\Rightarrow t^2 - 6t + 25 = 0, \text{ where } t = z^3$$

$$\Rightarrow t = \frac{6 \pm \sqrt{36 - 4 \times 1 \times 25}}{2}$$

$$\Rightarrow t = \frac{6 \pm \sqrt{36 - 100}}{2} = \frac{6 \pm \sqrt{-64}}{2}$$

$$\Rightarrow t = \frac{6 \pm 8i}{2}$$

$$\Rightarrow t = 3 \pm 4i$$

$$\Rightarrow t = 3 + 4i, 3 - 4i$$

$$\Rightarrow z^3 = 3 + 4i, 3 - 4i$$

$$\Rightarrow |z^3| = |3 + 4i|, |3 - 4i|$$

$$\Rightarrow |z|^3 = \sqrt{3^2 + 4^2}, \sqrt{3^2 + (-4)^2}$$

$$\Rightarrow |z|^3 = 5$$

$$\Rightarrow |z| = (5)^{\frac{1}{3}}$$

☒ 1 $(5)^{\frac{1}{3}}$

☐ 2 $(25)^{\frac{1}{3}}$

☐ 3 $(125)^{\frac{1}{3}}$

☐ 4 $(625)^{\frac{1}{3}}$

EXERCISE – 4.1

QUESTION

Express each of the complex number given in the Exercises 1 to 10 in the form $a + ib$.

$$(5i) \left(-\frac{3}{5}i \right)$$

$$z = \cancel{5i} \times -\frac{3i}{\cancel{5}}$$

$$\Rightarrow z = -3i^2 = -3 \times -1$$

$$\Rightarrow z = 3 = 3 + i \cdot 0.$$

Thank You

